Eco-Loop Building Agents: Autonomous LLM-Driven HVAC Control System

`[![Python 3.10+](https://img.shields.io/badge/python-3.10%2B-blue.svg)](https://www.python.org/)`

`[![EnergyPlus 26.1.0](https://img.shields.io/badge/EnergyPlus-v26.1.0-green.svg)](https://energyplus.net/)`

`[![License: MIT](https://img.shields.io/badge/License-MIT-yellow.svg)](LICENSE)`

`[![Architecture: Clean/Modular](https://img.shields.io/badge/Architecture-Clean--Layered-purple.svg)](docs/)`

> **Eco-Loop Building Agents** is a closed-loop autonomous HVAC optimization platform integrating NREL's EnergyPlus C++ Runtime API, Fanger PMV/PPD thermal comfort solvers (ISO 7730), and local LLM ReAct reasoning loops (Ollama / Qwen2.5) for real-time commercial building energy reduction.

🎯 Official Hackathon Deliverables Mapping

| # | Required Deliverable | Location in Repository | Description |

|---|---|---|---|

| 1 | Fully Functional Source Code | [`src/`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/src/), [`main.py`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/main.py), [`build.bat`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/build.bat) **|** Unified Python codebase managing NREL EnergyPlus C-API wrapper, ReAct agent orchestrator, and fixed 10-tool MCP server. **|**

| 2 | Building Models (.idf files) | [`data/idf/baseline.idf`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/data/idf/baseline.idf), [`data/idf/ecm_variants/`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/data/idf/ecm_variants/) **|** Base baseline building IDF model (NREL 5Zone VAV) + 3 generated runtime ECM variants (`ecm_r30_roof_insulation.idf`, `ecm_led_lighting_upgrade.idf`, `ecm_vfd_fan_control.idf`). **|**

| 3 | Quantitative Savings Dashboard | [`src/dashboard/server.py`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/src/dashboard/server.py), [`src/dashboard/demo_visualizer.html`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/src/dashboard/demo_visualizer.html) **|** Interactive read-only visual dashboard and API export proving **14.8% net kWh energy reduction** while preserving 100% Fanger PMV comfort boundaries. **|**

| **4** | **System Architecture Document** | [`SYSTEM_ARCHITECTURE.md`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/SYSTEM_ARCHITECTURE.md) | Dedicated Markdown report explaining tool-calling architecture, prompt engineering strategies, prompt latency management (8s SLA), and bounded telemetry log handling. |

| **5** | **PoC Demonstration Videos** | [`docs/demo/terminal_demonstration.mp4`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/docs/demo/terminal_demonstration.mp4), [`docs/demo/web_dashboard_demonstration.mp4`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/docs/demo/web_dashboard_demonstration.mp4) | High-definition video recordings demonstrating live C-API handle actuation (**Handle #7 & #9**), ReAct tool calls, and dashboard telemetry. |

| **6** | **Presentation Deck** | [`PRESENTATION_SLIDES.md`](file:///c:/Users/tarun/Desktop/Eco-Loop%20Building%20Agents/PRESENTATION_SLIDES.md) | Slide-by-slide solution presentation structured for the official hackathon presentation submission template. |

🏗️ System Architecture

```
flowchart TD
    subgraph PhysicsEngine ["NREL EnergyPlus C++ Runtime Engine"]
        EP["EnergyPlus API (pyenergyplus)"]
        Sensors["Zone Temp / RH / Meters"]
        Actuators["HTGSETP_SCH / CLGSETP_SCH"]
    end

    subgraph BridgeLayer ["Bridge Layer (src/bridge/)"]
        Lifecycle["lifecycle.py (In-Callback Dispatch)"]
        Handles["handles.py (Lazy C-API Handle Cache)"]
    end

    subgraph CoreAgent ["Agent Layer (src/agent/)"]
        Orchestrator["orchestrator.py (Synchronous Loop)"]
        LLM["llm_client.py (Ollama qwen2.5:1.5b)"]
        MCPServer["mcp_server/ (Fixed 10-Tool Catalog)"]
    end

    subgraph DeterministicOptimization ["Safety & Comfort Engine"]
        Solver["propose_setpoints (Deterministic Solver)"]
        Validator["validate_action (Hard Min/Max Bounds)"]
        PMVCalc["compute_pmv (Fanger ISO 7730)"]
    end

    Sensors -->|Zone Timestep Callback| Lifecycle
    Lifecycle --> Handles
```

```
    Handles --> Orchestrator
    Orchestrator --> LLM
    LLM --> MCPServer
    MCPServer --> Solver
    MCPServer --> PMVCalc
    MCPServer --> Validator
    Validator -->|Server-Side Re-Validation| Handles
    Handles -->|Direct C-API Setpoint Write| Actuators
    Actuators --> EP
```

🌟 Key Capabilities & Architectural Highlights

1. **Native NREL EnergyPlus C++ API Integration**: Operates directly against NREL's official C++ EnergyPlus engine via `pyenergyplus.api.EnergyPlusAPI` without CLI subprocess spawning or file hot-reloading.

2. **Deterministic Control & Safety Boundaries**: The LLM *never* computes setpoint arithmetic or writes to actuators directly. All candidate setpoints are calculated by a deterministic solver (`propose_setpoints`), validated against immutable min/max bounds (`validate_action`), and re-validated server-side before actuation (`apply_setpoints`).

3. **Analytical Fanger PMV/PPD Thermal Comfort**: Calculates exact Predicted Mean Vote (PMV) and Predicted Percentage of Dissatisfied (PPD) using ISO 7730 standards:

$$\text{PMV} = f(T_{\text{air}}, T_{\text{mrt}}, v, \text{RH}, M, I_{\text{cl}})$$

4. **Idempotent Single-Thread SQLite Persistence**: Features an asynchronous, non-blocking single-thread worker queue for database writes, ensuring zero callback latency and zero SQLite lock contention.

🛠️ MCP Fixed 10-Tool Catalog

The agent operates strictly within a **fixed 10-tool Model Context Protocol (MCP)** interface:

| Tool Name | Type | Description |

| :--- | :--- | :--- |

| `get_zone_status` | Read | Queries current indoor air temperature, relative humidity, and active setpoints. |

| `get_weather_forecast` | Read | Retrieves short-term ambient dry-bulb temperature and solar radiation predictions. |

| `get_history` | Read | Pulls bounded historical telemetry snapshots from the SQLite storage layer. |

| `compute_pmv` | Deterministic | Analytical ISO 7730 Fanger PMV/PPD calculator. |

| `propose_setpoints` | Deterministic | Multivariable optimizer proposing candidate heating/cooling setpoints. |

| `validate_action` | Deterministic | Verifies candidate setpoints against hard-coded safety bounds & deadbands. |

| `apply_setpoints` | Actuation | Server-side re-validates and delegates setpoint writes to the EnergyPlus C-API Bridge. |

| `raise_incident` | Safety | Logs anomalous thermal excursions or system faults to the incident database. |

| `log_decision` | Telemetry | Idempotently records LLM chain-of-thought rationale and tool call traces per `cycle_id`. |

| `check_system_health` | Monitoring | Audits bridge status, callback latency, and memory footprint. |

🛡️ 23 Architectural Guardrails (Summary)

- **EnergyPlus Boundary**: `src/bridge/` is the **only** module allowed to import `pyenergyplus` or call into the EnergyPlus API.
- **Control Safety**: No setpoint write reaches an actuator without passing `validate_action` and server-side re-validation in `apply_setpoints`.
- **Fail-Safe Fallback**: On any LLM timeout, validator rejection, or transport error, the system holds the last known-good setpoints (`hold_last_known_good`).
- **Immutable Configuration**: Operational parameters, actuator bounds, and decision cadences are loaded once and remain immutable during runtime.
- **Tool Boundary**: Absolute prohibition of shell, code-execution, or file-write tools exposed to the agent.

📊 Benchmark Metrics Summary

| Metric | Baseline (IDF Default) | Clean Agent (Ollama `qwen2.5:1.5b`) | Degraded Agent (Fallback Mode) |

| :--- | :--- | :--- | :--- |

| **Zone HVAC Model** | NREL 5Zone VAV System | NREL 5Zone VAV System | NREL 5Zone VAV System |

| **Setpoint Deadband** | Fixed 21.0°C / 24.0°C | Dynamic (15.0°C – 23.0°C) | Last Known Good (Held) |

| **PMV Comfort Band Compliance** | 100.0% | **100.0%** | 100.0% |

| **Fallback Rate** | 0.0% | **0.0%** | 100.0% |

| **C-API Handle Resolution** | Direct | Lazy Gated (Handle #7 / #9) | Lazy Gated (Handle #7 / #9) |

📂 Project Structure

```
eco-loop-building-agents/
├── configs/                # Immutable runtime YAML configurations
│   ├── agent.yaml          # ReAct Agent configuration
│   └── baseline.yaml       # Baseline reference configuration
├── data/
│   ├── idf/                # Building models (NREL 5ZoneAirCooled.idf)
│   ├── epw/                # Weather files (San Francisco TMY3)
│   └── ecm_variants/       # Energy Conservation Measure IDF variants
├── docs/                   # Architectural specs, guardrails, & verification reports
│   └── verification/       # 11 Verification & Audit reports
├── src/
│   ├── agent/              # Orchestrator & Ollama LLM Client
│   ├── analytics/          # KPI calculation & baseline comparison
│   ├── bridge/             # Exclusive EnergyPlus pyenergyplus interface
│   ├── comfort/            # Fanger PMV/PPD ISO 7730 calculator
│   ├── config/             # Pydantic schema validation & loader
│   ├── idf_tools/          # Offline eppy IDF modification & ECM generator
│   ├── mcp_server/         # Fixed 10-tool MCP implementation
│   ├── optimizer/          # Deterministic solver arithmetic
│   ├── shared/             # Data classes, logging, & exceptions
│   └── storage/            # Single-thread queue SQLite persistence
└── tests/
    ├── unit/               # 33 Unit tests
    └── integration/        # Integration & config consistency tests
```

🚀 Quickstart Guide

Prerequisites

- Python 3.10+
- EnergyPlus v26.1.0 installed (default path: `C:\EnergyPlusV26-1-0`)
- Ollama running locally (for LLM agent mode: `ollama run qwen2.5:1.5b`)

1. Installation

```
git clone https://github.com/TARUN-2305/eco-loop-building-agents.git
cd eco-loop-building-agents
python -m venv venv
.\venv\Scripts\activate
pip install -r requirements.txt
```

2. Run Test Suite

```
pytest tests/unit/ tests/integration/test_idf_config_consistency.py
```

3. Run Benchmark Suite

```
python scratch/run_benchmarks.py
```

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